

Symptom-based Diabetes Classification and Unsupervised Type Profiling

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Abstract

Driven by the critical clinical importance of the early detection of type 1 diabetes, this project proposes a multi-stage machine learning framework using a symptom-based clinical dataset to address the absence of explicit disease-type labels.

Utilising data from 520 patients collected at the Sylhet Diabetes Hospital, which tracks 16 categorical and numerical attributes (including age, gender, polyuria, polydipsia, and sudden weight loss) mapped to a binary target, the methodology establishes a dual-layer approach.

In the first stage, a supervised binary classification pipeline evaluates tabular classifiers—such as Decision Trees and Random Forests—to accurately distinguish between healthy and diabetic cohorts, mathematically prioritising classification over regression to target discrete diagnostic labels.

In the second stage, an unsupervised clustering layer utilising K-Means ($k=2$) is applied strictly to the predicted positive instances. This allows for an empirical evaluation of whether latent clinical signatures naturally partition into two distinct clusters aligning with known epidemiological profiles of type 1 (acute, early-onset) and type 2 diabetes.

To ensure methodological rigor, data preparation is managed via an end-to-end scikit-learn transformer pipeline to prevent data leakage

during categorical encoding and feature scaling, while stratified k-fold cross-validation and systematic hyperparameter tuning are implemented to mitigate overfitting risks on the 520-instance sample.

1 Introduction

Diabetes mellitus represents one of the most pervasive global health challenges of the modern era, characterised by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both. Early and accurate detection of the condition is essential to mitigate severe long-term complications, including retinopathy, nephropathy, neuropathy, and cardiovascular disease. However, from a clinical management and epidemiological tracking perspective, simply diagnosing diabetes is insufficient; identifying the specific disease etiology—primarily distinguishing between Type 1 Diabetes Mellitus (T1DM), driven by autoimmune beta-cell destruction, and Type 2 Diabetes Mellitus (T2DM), driven by progressive insulin resistance—is critical for defining appropriate therapeutic pathways.

A primary challenge in utilising public clinical data for machine learning research is the frequent absence of explicit subtype labels. Public repositories, such as the symptom-based dataset collected from the Sylhet Diabetes Hospital, typ-

ically capture diagnostic outcomes as a simplistic binary classification task (Positive/Negative). While highly valuable for initial screening, this constraint leaves the underlying subpopulation structures hidden. This project addresses this explicit limitation by designing and implementing a robust dual-stage machine learning framework. Level 1 establishes a highly accurate supervised classifier to separate healthy individuals from diabetic patients. Level 2 subsequently isolates the predicted positive cohort and applies unsupervised learning to discover whether the underlying clinical signatures naturally partition into groups matching known T1DM and T2DM clinical proxies.

2 Proposed Method

The architectural design of the proposed multi-stage hybrid pipeline strictly follows the standardised CRISP-DM (Cross-Industry Standard Process for Data Mining) lifecycle. The framework is split into two primary computational layers: a Supervised Binary Classification layer (Level 1) and an Unsupervised Cohort Stratification layer (Level 2), bound together by a leakage-free pre-processing pipeline.

2.1 Data Pre-processing and Leakage Prevention

The dataset consists of clinical questionnaire records for 520 patients tracking 16 primary attributes: one continuous feature (*Age*), one demographic feature (*Gender*), and 14 binary clinical signs (e.g., *Polyuria*, *Polydipsia*, *Sudden weight loss*, *Obesity*) mapped to the target attribute *Class*.

To maintain clinical validity, this analysis deviates from traditional data cleaning practices, prioritising the preservation of patient diversity over standard statistical homogenisation:

- **Retention of Outliers:** Statistical outlier removal (e.g., Z-scores or IQR) was omitted for the ‘Age’ attribute. Extremely young

and old patients are retained as valid clinical profiles, preventing potential bias against atypical diagnostic cases.

- **Dataset Completeness:** As the source dataset contained no null values across the 520 records, no imputation or deletion strategies were required.
- **Preservation of Categorical Diversity:** Given that features are self-reported binary symptoms, variability in responses is interpreted as genuine clinical diversity rather than data noise. By treating this variance as phenotypic variation, the clustering phase remains sensitive to the underlying clinical signatures of individual patients.

To enforce absolute statistical integrity, a Stratified Train-Test Split (80/20 ratio) is executed before any pre-processing operations or parameter estimations occur. This ensures that validation matrices reflect true generalisation capabilities and completely eliminates data leakage. Following the split, a centralised **ColumnTransformer** handles the heterogeneous data types automatically:

1. **Categorical Encoding:** The binary textual features and gender flags are mapped to numeric spaces using a **OneHotEncoder** with the parameter `drop='first'` to avoid multicollinearity.
2. **Continuous Scaling:** Because the Level 2 clustering mechanism relies heavily on distance metrics, **MinMaxScaler** was intentionally selected to transform the *Age* attribute, scaling it strictly into the interval $[0, 1]$. This establishes absolute mathematical parity with the one-hot encoded binary features and prevents *Age* from dominating distance functions.

2.2 Level 1: Supervised Binary Classification

Level 1 focuses on separating healthy subjects from diabetic cases using tabular ensemble clas-

sifiers. The **Random Forest classifier** was selected for the primary diagnostic screening phase. Its ensemble nature effectively manages complex, non-linear symptom interactions, offering greater stability and reliability than single decision trees, which are prone to overfitting. The interpretability provided by 'Feature Importance' metrics further supports clinical auditability, clarifying the symptomatic drivers of classification.

To optimise model convergence, hyperparameter optimisation is handled via a grid search engine combined with a 5-fold **StratifiedKFold** cross-validation strategy. The cross-validation framework optimises the model based on the F_1 -score, ensuring an equal balance between precision and recall across folds.

2.3 Level 2: Unsupervised Cohort Discovery

Once the Level 1 system achieves a robust validation profile, it processes the entire dataset. The framework extracts all instances predicted as *Positive* (diabetic) to form the independent Level 2 clustering cohort.

K-Means clustering was chosen for the type discovery. This choice was driven by:

- **Clinical Actionability:** Its centroid-based approach allows for the clear definition of clinical 'centers of gravity'. It was decided to use one approach focused on Type 1 Diabetes and another for Type 2 phenotypes.
- **Exhaustive Stratification:** K-Means guarantees that every patient is assigned a group, which is crucial for subsequent triage and the identification of 'Atypical/Mixed' cohorts through consensus analysis.
- **Binary Feature Stability:** It provides robust partitions in discrete binary feature spaces, where density-based algorithms like DBSCAN often struggle to produce interpretable results due to sensitivity to parameter settings.

The binary symptoms were treated in 2 ways: firstly, for both types of diabetes, all symptoms were treated equally to then apply a weighting function to emphasise "anchor symptoms" (e.g., Polyuria for Type 1, Obesity for Type 2). This modification to the distance matrix directs the K-Means algorithm to prioritise medically significant markers, effectively refining the cluster centroids to better align with established clinical phenotypes. Then further analyses were conducted to check if increasing the value of k or applying feature engineering or pruning led to meaningful improvements to the results.

3 Results and Discussion

The experimental validation of the multi-stage pipeline produced clear quantitative metrics that validate both the classification and cohort discovery components.

3.1 Level 1 Classification Performance

The optimised Random Forest classifier achieved good generalisation metrics on the unseen test partition. The final parameters selected were `n_estimators=150`, `max_depth=8`, and `min_samples_split=5`. (Check table 1)

The model achieved an overall accuracy of 97%, with an F_1 -score of 0.98 for the positive class. The confusion matrix indicated that only 2 false negatives and 1 false positive were generated out of the 104 test instances.

To establish model interpretability and satisfy the requirement for explainable AI, feature importance analysis was conducted on the optimised Random Forest classifier. The analysis revealed that clinical symptoms—specifically *Polyuria* and *Polydipsia*—were the dominant predictors for diabetes screening, significantly outweighing demographic factors. Furthermore, an ablation experiment was performed by removing the *Age* and *Gender* features from the dataset. The model maintained its near-

perfect accuracy without these demographic inputs, mathematically proving that the isolated clinical symptom profile is sufficient for robust Level 1 screening.

Level 2: Iterative Phenotype Subtyping and Experimental Sequence

The second stage of the framework focused on differentiating clinical phenotypes within the diabetic cohort ($N = 320$). An exhaustive experimental sequence was conducted to identify the optimal balance between mathematical cluster separation and clinical diagnostic consensus across the approaches.

Phase I: Baseline and Weighted Initializations

Analysis began with a baseline **Unweighted K-means** ($k = 2$), which yielded a moderate agreement rate of 50.6%. To account for the clinical significance of specific symptoms like Polyuria, there was a transition to **Weighted K-means** ($k = 2$). While weighting improved the Silhouette score to 0.49, it did worsen consensus. Expanding these weighted models to $k = 3$ provided further evidence of the "Consensus Paradox": geometric separation improved, but clinical interpretability began to degrade as the model attempted to isolate a "mixed" symptom group.

Phase II: Feature Engineering and Pruning

To resolve the overlap between clusters, **Feature Engineering** was applied, incorporating the Hyperglycemic Crisis Score (HCS) and Metabolic Syndrome Proxy (MSP). These models were tested for both $k = 2$ and $k = 3$. The $k = 3$ engineered model reached a peak Silhouette score of 0.545, yet diagnostic consensus collapsed to 7.81%, indicating that complex features were capturing latent noise rather than clinical reality. Subsequent **Feature Pruning** ($k = 2, 3$) was performed to remove low-variance

symptoms; however, this resulted in a loss of granular detail without a compensatory gain in consensus stability.

Phase III: Tiered Age-Binning Strategy

The final iterations utilized **Age-Tiering**, discretizing the cohort into biological age bins to ground the clusters in clinical context. Both **Weighted and Unweighted Tiered** models have been evaluated for $k = 2$ and $k = 3$. The **Unweighted Tiered** ($k = 2$) model emerged as the optimal configuration, achieving the study's highest clinical consensus rate of 75.62%. This confirmed that age-based stratification is more effective for diagnostic triage than high-dimensional feature engineering or increased cluster counts.

4 Conclusion

This report successfully developed and validated a dual-stage machine learning framework for diabetic subtyping. Level 1 established a high-performance Random Forest pipeline (97-99% accuracy) using patient-reported symptoms, identifying Polyuria and Polydipsia as the primary diagnostic proxies.

Level 2 exploration revealed a significant "Consensus Paradox": while advanced feature engineering (HCS/MSP indices) and $k = 3$ initializations improved mathematical cohesion (Silhouette 0.54), they simultaneously degraded clinical diagnostic agreement to as low as 7.8%. The study concluded that a Tiered Age-Binning strategy was instrumental in resolving this ambiguity, in a way closer to the baseline approach though with a higher silhouette score. The Unweighted Tiered ($k = 2$) model emerged as the most robust solution for high-volume triage, achieving a 75.6% clinical consensus rate, whereas the Weighted Tiered approach served as the optimal middle-ground for specialist research with a Silhouette score of 0.530.

Future work should integrate physiological biomarkers such as HbA1c, C-peptide levels, Ketone levels, and BMI. These markers would likely resolve the remaining "Consensus Paradox" by providing the objective tie-breakers that symptom-only models lack, transitioning the framework from a symptomatic triage proxy to a definitive clinical diagnostic tool.

References

- [1] Arsha Prasad (2021) *Diabetes Dataset*, Kaggle. (Accessed: 2026-07-13) <https://www.kaggle.com/datasets/arshaprasad/diabetes-dataset>.

Table 1: Level 1 Classification Performance Comparison

Model / Class	Precision	Recall	F_1 -Score	Support
<i>Decision Tree</i>				
Negative (Healthy)	0.98	1.00	0.99	40
Positive (Diabetic)	1.00	0.98	0.99	64
Accuracy			0.99	104
<i>Random Forest</i>				
Negative (Healthy)	0.97	0.97	0.96	40
Positive (Diabetic)	0.98	0.98	0.98	64
Accuracy			0.98	104

Table 2: Subtyping Strategy Performance: Geometric Quality vs. Clinical Consensus

Strategy	Silhouette Score	Agreement Rate (%)	Clinical Utility
Unweighted Baseline	0.323	50.60	Initial Triage
Weighted $k = 3$	0.545	7.81	Latent Research
Engineered (HCS/MSP)	0.519	31.88	High-Sensitivity
Unweighted Tiered	0.302	75.62	Optimal Triage
Weighted Tiered	0.530	49.38	Clinical Profiling